

Boughton Bay

Shoreline length: 68 km



Shore Types

Boughton Bay SMU includes an open coastline facing East and an estuary protected by a barrier sandspit, with a majority of low plains and wetlands.

% Shore types within SMU

low plain	bluff	sloped armour	cliff	wetland	vertical struct.	dunes	sandspit
57	12	2	3	21	2	2	1



Annandale Wharf and barrier sandspit at entrance of Boughton Bay



Transition from updrift eroding bluff to dunes



Segmented armouring along outer bluff



Partial armouring along Harbour road -



Bluffs inside the bay



Honey Bee Lane on barrier dune to MacPhersons Pond



Launching Pond harbour

Governance, Infrastructure, and Economy

Boughton Bay is largely covered by three jurisdictions: the Rural Municipality of Annandale-Little Pond-Howe Bay and the Rural Municipality of Central Kings (~46% combined), an unincorporated area (~16%), and the Town of Three Rivers (~37%).

The area contains a large number of roads and facilities, including marine infrastructure at Launching Pond; Harbour Road to Launching Pond Wharf; HoneyBee Lane (Launching, at the beach at MacPhersons Pond); Beach Road (Launching, at the beach); Long Wharf Road (Launching, at the beach); McCarron and Montreal Roads (St. Georges, at the beach); Chapel Point Road (at the beach, including a breakwater, St. Georges); Wharf Road at Poplar Point; Lighthouse Road (Annandale, at the beach); Nortons Road (Annandale, across a wetland, to Annandale wharf); Tassell Road (Annandale); Annandale Road (Route 310) over Blackett's Creek; Route 4 -Seven Mile Road, Route 311 - Primrose Road, and Spry Point Road (Little Pond, at the beach).

Coastal economic drivers that are also infrastructure include Annandale Marina (wharf), the wharf at Launching Pond, New Port Wharf, and Cardigan Wharf.

Local and legacy land use constraints include several subdivisions with small lots on unpaved roads running parallel or perpendicular to the coast.

Natural Assets and Heritage

Parks and cultural facilities include the Annandale Front Range Lighthouse (historical landmark), the Annandale Rear Range Light (historical landmark, Nortons Road), Central Kings Park (playground), Bridgetown Ballfield, and the Boughton River Natural Area (hiking trails).

The shoreline includes critical habitat for bird species including the endangered Piping Plover and threatened Bank Swallow.

Planning and Policy Framework

Three Rivers has municipal planning authority over more than a third of this SMU. Development on the coast is informed by the Environmental Protection Act, including a 15m buffer, with no additional building setbacks aside from the buffer. The Official Plan includes specific guidance permitting armouring, though such provisions are secondary to any provincial standards, and includes some policies regarding expansion of buildings within the sea level rise overlay. There are no specific flood elevations or vertical setbacks established, but the Official Plan does include specific erosion-based setbacks, with the possibility of reduced setbacks under certain conditions. Hazards are assessed at the subdivision stage.

Elsewhere in this SMU (Annandale-Little Pond-Howe Bay, Central Kings, and the unincorporated portion), provincial rules apply. Land use and development is managed under provincial development standards,

which include horizontal coastal building setbacks based on edge of feature and a calculation of erosion over time, building setbacks for dunes, and a requirement for a coastal buffer for new subdivisions.

There are no specific flood elevations identified in regulations, but coastal hazard assessments are typically conducted.

Coastal Flood and Erosion Risk

While low plains and wetland shores are vulnerable to increasing flood hazards, infrastructure risk is low. The erosion hazard is highest along the exposed east shoreline. Erosion hazard inside the estuary is limited by the sheltered exposure and barrier sandspit.

The following flood risk indicator is based on the even with 1% annual exceedance probability (100-year return) with increasing amounts of sea level rise (SLR). The erosion risk indicator is based on potential shoreline change or salt marsh migration to future time horizons.

Number of at-risk buildings [N. per km shoreline]

	Present	0.3m SLR (2050)	0.6m SLR (2080)	1.2m SLR (2130)
SMU Boughton Bay				
Flood risk	85 [1/km]	88 [1/km]	95 [1/km]	110 [2/km]
Erosion risk	N/A	48 [1/km]	58 [1/km]	127 [2/km]
PU Launching Pond Harbour				
Flood risk	48 [23/km]	48 [23/km]	49 [24/km]	49 [24/km]
Erosion risk	NA	16 [8/km]	18 [9/km]	37 [18/km]
PU Annandale Wharf				
Flood risk	18 [21/km]	18 [21/km]	21 [25/km]	22 [26/km]
Erosion risk	5 [6/km]	5 [6/km]	24 [29/km]	5 [6/km]

Sediment transport - Sediment transport is highest along the exposed seaside including cliffs, bluffs, dunes, barriers, and wetlands. Sediment transport north of Launching Pond Harbour feeds the barrier sandspit at the entrance of Boughton Bay.

Proposed SMP Strategies

Given the initially low risk levels the proposed strategy is NAI, which will give time to prepare for MR. In PUs where critical infrastructure is at risk, resist strategies are initially acceptable until the flood risk is such that MR would be warranted.

	Short-term	Medium-term	Long-term
Sea level rise range	0 – 0.3 m	0.3 – 0.6 m	0.6 – 1.2 m
Potential timeframe	2030 - 2050	2050 - 2080	2080 - 2130
Proposed SMU strategy	NAI	MR	MR
Rationale	Mid- to long-term MR will reduce risk to the limited number of vulnerable assets while preserving natural shorelines. Notably, the Old Ferry sand spit protects Boughton Bay from wave action and needs natural input from the south, which would be negatively affected by armouring.		
Proposed PU strategy			
Launching Pond Harbour	R	R	R
Rationale	A Resist strategy that includes nourishment is warranted for the south barrier beach to mitigate the risk of breaches.		
Proposed PU strategy			
Annandale Wharf	R	R	MR
Rationale	A Resist strategy can maintain this critical infrastructure until the area is at risk of being flood isolated. MR (raise land or retreat) is advised past the end of the existing infrastructure lifetime.		



LEGEND

- Policy Unit
- Shoreline Management Unit
- Net Longshore Sediment Transport
- High Hazard Area
- Flood Extent - Present Day (100-year Return)
- Flood Extent - 1.2m SLR (100-year Return)
- Property Boundary
- Building > 90ft²

0 375 750 1,500
Metres

Coordinate System:
NAD 83 CSRS PEI Double Stereographic

NOTES:

**COASTAL EROSION
AND FLOOD HAZARD**

Shoreline Management Unit:
Boughton Bay

Prince Edward Island
Île-du-Prince-Édouard